Code availability is not enough: A schema-based representation for comprehensible and extensible data analysis

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ABSTRACT
Current reproducibility practices have greatly improved our ability to replicate published results, but they do not necessarily make data analyses easier to understand. Audience seeking to comprehend an analysis need to piece together information scattered across manuscripts and code scripts to understand the sequence of analytical steps and the rationale behind them, with some details may not explicitly record in either source. Exploring alternative analytical choices is even more challenging, as it requires understanding and modifying code that may be unfamiliar and consider how it would impact upstream and downstream code. To address these challenges, we propose a schema that represents data analyses as a collection of analytical decisions, together with their justification, inputs, and outputs. Coupled with large language models (LLMs), the schema can be inferred from natural language descriptions and translated into analysis scripts. The structured representation also enables exploration of alternative analytical choices, allowing multiple analytical workflows to be generated and compared. We demonstrate the proposed workflow using the many-analysts football study, where LLMs create the schema from the decision choices in team reports, generate corresponding analysis scripts, and explore alternative random effect specifications. As an intermediate representation between the manuscript and code scripts, our schema makes analyses more comprehensible and extendable from a reader's perspective.

KEYWORDS
Reproducibility; Large Language Models (LLMs); Data analysis

1. Introduction

The reproducibility movement has been a major development in modern scientific research, motivated in part by concerns regarding the replication of published findings and the increasing reliance on computational methods across disciplines. In statistics, advocates of reproducible research argue that scientific claims should be accompanied by the data and code necessary to regenerate reported results (Peng, 2011). These

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practices have greatly improved the reproducibility of published results across scientific domains (Kohler et al., 2026; Xu & Yang, 2026) and have been incorporated into journal policies through data and code availability requirements (see the *Journal of the American Statistical Association* editorial (Wrobel et al., 2024)).

An implicit assumption underlying current practices is that reproducibility implies understandability, given that the data and the code required to reproduce the results are available to readers. Specifically, understandable implies that an independent researcher should be able to, without much difficulty, track the decisions made by the authors, understand the rationale behind those decisions, and comfortably modify the code to explore alternative choices. As data analysis grown in complexity in modern applied research, understanding a piece of analysis has become non-trivial. Decisions related to data preprocessing and exploratory data analysis are often reported only briefly in manuscripts and left as technical details for readers to infer from supplementary materials. Many decisions are not made linearly but emerge through an iterative process (Jun, Birchfield, et al., 2022; Jun, Seo, et al., 2022; Zhang & Peng, 2025), and it is not entirely clear how this iterative nature of data analysis can be communicated. Making changes to an existing script can often be difficult because analytical decisions are intertwined with implementation details, project-specific abstractions, and dependencies across different parts of the workflow, requiring readers to consider how modifications propagate through upstream and downstream components. As a result, readers seeking to understand, modify, or extend an analysis must often spend a non-trivial amount of the time to piece together information scattered across manuscripts, code scripts, and supplementary materials to fully understand an analysis.

Advances in Large Language Models (LLMs) have led to widespread use of these systems for code generation tasks, including code completion and program repair (Jiang et al., 2026; Shah et al., 2026). Early work on code-oriented LLMs demonstrated strong capabilities in translating natural language descriptions into executable programs, with models such as Codex achieving substantial improvements on benchmark programming tasks (Chen et al., 2021; Huang et al., 2022; Lai et al., 2023). These capabilities have led to broad adoption of coding assistants through web interfaces, APIs and integrated development environments. In the R ecosystem, packages such as `ellmer` (Wickham et al., 2025), `chatLLM` (Owusu Boakye, 2026a), and `LLMAgentR` (Owusu Boakye, 2026b) provides increasingly accessible interfaces for access to latest LLM models.

Building on these advances, recent work explores fully automated systems that use multiple LLM agents to plan, generate, execute, and refine data analyses (Hu et al., 2024; Jansen et al., 2025). Our work relies on the same underlying capability of translating analytical specifications into code, but differs in its view of what should be automated. Data analysis is often underdetermined, with many plausible choices regarding data preprocessing, transformation, aggregation, and model specification (Bertran et al., 2026; Gao & Xiao, 2026; Luo et al., 2025), by contrast, once these analytical decisions have been specified, translating them into executable code is often a comparatively straightforward task. This distinction motivates our focus on representing analytical

decisions explicitly and using LLMs to translate those decisions into code.

In this paper, we present a schema designed to describe data analysis in a structured way that captures the decisions and their justifications, as well as the inputs and outputs of each step. We also demonstrate how this schema, coupled with LLMs, can be used to generate code scripts from natural-language descriptions in manuscripts, and explore alternative decision choices. We apply our schema to the many-analyst football study, where LLMs create the schema from the decision choices reported in the final reports of each team, and generate and validate the corresponding code scripts against the original published results. We further demonstrate how to the schema supports specification and exploration of alternatives random effect specification for Team 5.

The paper is organized as follows: In Section 2, we present our methodology, including the schema representation for describing data analysis and the workflow for generating and validating code from the schema by LLMs. In Section 3, we apply our methodology to the many-analyst football and demonstrate that we can reproduce the results reported in the paper, identify errors in the original reports, and highlight failure modes of current LLMs. Finally, in Section 4, we discuss the implications of our work and potential future directions.

2. Methodology

As traditional reproducibility practices can be insufficient for understanding complex modern data analyses, we have designed two complementary YAML schemas and the associated workflow to better support comprehension and extensibility in applied data analysis. Our workflow is designed to be 1) decision-centric: analyses are represented as a sequence of decisions with explicit rationale, and 2) reader-oriented exploration: readers can inspect, compare, and extend those decisions.

2.1. *A schema for describing data analysis*

We represent a data analysis, $\mathcal{S} = \{A_1, A_2, \cdots, A_n\}$, as a sequence of n analytical step A_i, applied to a dataset $\mathcal{D}$, where each step, A_i, is defined as a tuple:

$$A_i = \langle f_i, \pi_i, r_i, \mathcal{I}_i, \mathcal{O}_i, \omega_i \rangle. \tag{1}$$

The components of A_i can be partitioned into two parts: a required triplet (f_i, π_i, r_i) and an optional triplet $(\mathcal{I}_i, \mathcal{O}_i, \omega_i)$. The first component of the triplet, $f_i \in \mathcal{F}$ is the *objective* of a step. The second component, $\pi_i \in \Pi$, represents the specific *decision* made to achieve the objective f_i and the term r_i is the *justification* provided for choosing the decision π_i. Structuring this information into a schema involves defining the three components described above for each step. While the decision (what has been done) and

the justification (why it is done) are generally clear, the objective requires identifying what the step is trying to achieve. In other words, it represents the broader goal that could potentially be fulfilled through other alternative choices (formally introduced in Section 2.2). As an example, consider the description for calculating the Human Development Index (HDI) (United Nations Development Programme, 2025):

1) Scale all four variables (life expectancy, expected years of schooling, mean years of schooling, and GNI per capita) to the [0, 1] range.
2) Combine the two education variables (expected years of schooling and mean years of schooling) into a single education variable.
3) Combine the three dimensions (health, education, and income) into a single index using a geometric mean.

In Step 3, the objective is to combine the three dimensions. While a simple arithmetic mean could be an alternative decision to fulfill this goal, the geometric mean (the decision) is chosen because it penalizes uneven development among the three dimensions (the justification). The full schema for constructing HDI is presented in Listing 1.

Listing 1 An example schema for calculating Human Development Index (HDI)

```
meta:
  type: schema
nodes:
- tag: step-scaling
  objective: how to handle variables in different scales?
  decision: apply min-max scaling to each variable
  rationale: to convert the variables to the same scale for future
    construction of a single index
- tag: step-education
  objective: how to combine the two education-related variables?
  decision: take the average
  rationale: the arithmetic mean allows perfect substitutability
    between the two variables
- tag: step-combine
  objective: how to combine the three dimensions into a single index?
  decision: use the geometric mean
  rationale: the geometric mean penalizes uneven development
    among the three core dimensions
```

On the optional components of a data analysis step A_i, the operational flow of the analysis is governed by the input and output spaces, $\mathcal{I}_i$ and $\mathcal{O}_i$ respectively, once a dataset is provided. While the output of a step can be a pseudo object that not currently exist in the dataset, the inputs for any given step must be within the existing data scope, comprising variables in the original dataset and all outputs from previous steps. Formally, we require:

$$\mathcal{I}_i \subseteq \left(\mathcal{D} \cup \bigcup_{j<i} \mathcal{O}_j \right), \quad \forall i \in \{2, \ldots, n\}.$$

When the schema is extracted automatically by Large Language Models (LLMs) from text, the inputs and outputs will be inferred by the LLMs, if a dataset is provided. We also introduce a categorical confidence score, ω_i (low, medium, high) that can be useful in case when the source text is ambiguous to guide the LLM to ask for clarification to consolidate the schema extraction.

2.2. *Describing alternatives*

At each step, one decision choice is made by the researchers to address an objective, however the choice is often not the only possible one. For step i, there could be k_i decision choices that fulfill the objective f_i: $\{A_i^1, A_i^2, \ldots, A_i^{k_i}\}$. We can define a selection vector $r = (r_1, r_2, \ldots, r_n)$, where $r_i \in \{1, \ldots, k_i\}$, to describe an analysis comprised of specific series of decisions: $\mathcal{S}^r = \{A_1^{r_1}, A_2^{r_2}, \ldots, A_n^{r_n}\}$. A multiverse, used to describe the entire decision space, is defined as the Cartesian product of these sets: $\mathcal{M} = \{\mathcal{S}^r \mid r_i \in 1, \ldots, k_i, \forall i\}$. The cardinality of this space, $|\mathcal{M}| = \prod_{i=1}^{n} k_i$, quantifies the total number of unique, fully-specified analysis.

Returning to the HDI example, in Step 3, instead of a geometric mean, an arithmetic mean is another possibility to combine the three dimensions into a single index. It may not be the best decision, but still a plausible one to derive a single index from the three dimensions, hence achieve the objective of this step. Listing 2 shows the structure of an alternative YAML file. The `meta` section identifies the file type (an alternative file rather than a schema file) and specifies the analysis step and the objective under consideration. Individual alternatives are defined as entries in the alternatives list, allowing multiple alternatives to be specified within a single file.

Listing 2 An example YAML file for specifying alternative for Step 3 in calculating Human Development Index (HDI).

```
meta:
  type: alternative
  step: step-combine
  objective: how to combine the three dimensions into a single index?
alternatives:
  - id: arith-mean
    decision: use an arithmetic mean
    rationale: a simple average of the three dimensions
```

2.3. *Workflow*

The overall workflow for schema-based data analysis is illustrated in Figure 1. A schema can be created either manually from the template or automatically by an LLM from a natural-language manuscript. Once created, the schema serves as a structured specification that guides a second LLM in generating the code for the analysis. To improve reliability, our design includes a third LLM agent which is responsible for code validation. While the latest LLMs can generate reasonable code based on prompt, they do not consistently achieve perfect accuracy. The validation process runs the generated script in a sandbox environment and if an error occurs, a third LLM agent will be employed to diagnose the script. The validation agent receives the generated code together with the error message and the corresponding line number produced during the execution, and returns a newer version of the script. The revised script is then re-executed in the sandbox and this cycle continues until one of the three conditions is met: the code runs successfully, a maximum number of attempt is reached, or the user terminates the process. As will be show in Section 3, this validation process resolves the majority of R syntax issues encountered during data wrangling and statistical modelling tasks. On top of supporting a single analytical specification, the workflow also allows users to argument a schema with an alternative YAML file. This expands the original schema into a multiverse of parallel schemas, each representing a different sequence of decision choices. The same code generation and validation workflow is then applied independently to each schema, producing a corresponding set of validation scripts.

Returning to the HDI example, the proposed workflow allows both authors and readers to interact with a data analysis through natural language. The schema in Listing 1 is straightforward to construct from a template and can be included as supplementary material to document the analytical workflow. An author may use the "code generation and validation" path in Figure 1 to automatically generate the analysis script for computing HDI and validate its accuracy. A reader, on the other hand, may wish to explore how the HDI values or country rankings change if the original arithmetic mean (instead of the geometry mean) is used. By combining the original schema in Listing 1 with the alternative specification Listing 2, the "alternative generation" path produces a corresponding analysis script for each alternative with LLMs. In contrast, modifying an existing code script by hand typically requires manually coordinating changes across upstream and downstream code. By representing analyses as a sequence of structured analytical decisions, our workflow allows LLMs to automate much of this process, making it easier to understand the sequence of steps taken in the analysis and to explore alternatives.

3. Application: many-analysts football study

In this section, we apply our methodology to the many-analyst football by Silberzahn et al. (2018) to reproduce the reported estimated effect size by each team. During this process, we identify some errors in the original reports and highlight failure modes of

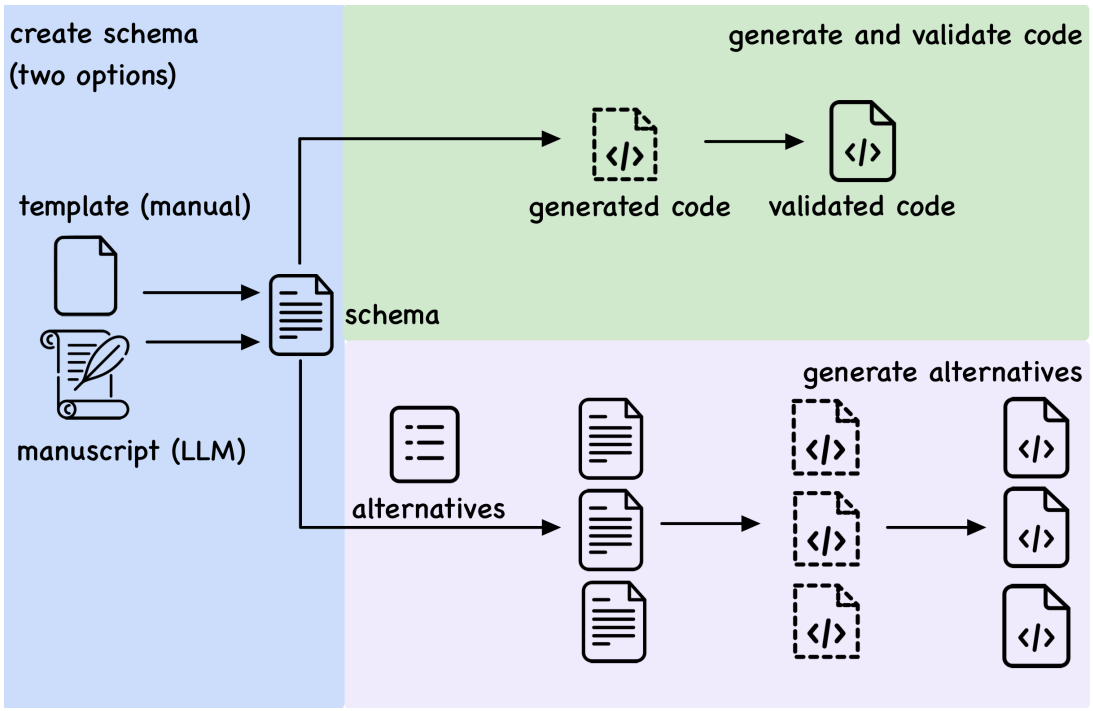


Figure 1. Overview of the schema-driven analysis workflow. Schemas can be authored manually or generated from natural language manuscripts by an LLM. Schemas are used to generate code, which is validated through iterative execution in a sandboxed environment and repaired when errors occur. Alternative specifications can expand a schema into a multiverse of parallel analyses.

current LLMs.

Silberzahn et al. (2018) invites 29 research groups to conduct the same data analysis to answer whether are more likely to give red cards to dark-skin-toned player than to light-skin-toned players. The provided dataset contains two raters' skin color rating to players' picture, the aggregated number of red card received by a player from a referee across all matches in that player-referee pair, and other relevant player, referee, and game characteristics: player birthday, height, weight, player position, league country, number of victories, ties, and defeats, and number of yellow and red cards received. In Silberzahn et al. (2018), each group first conducted an initial analysis separately, followed by a round-robin peer evaluation. Each group was then given the opportunity to incorporate feedback from the discussion and submitted a final analysis as a report with a formal Method section. All the materials related to the study is publicly available on the Open Science Framework (OSF).

Based on the reports presented by each team, most participating groups adopted a modelling approach to estimate the effect of skin color rating on the likelihood of receiving a red card. As an illustrative example, Group 5 first transformed the data into player–game observations, resulting in a binary response variable for whether a player received a red card, and removed observations with missing values using list-wise deletion. The resulting data were modelled using a generalized linear mixed model (GLMM) with a binomial error distribution. The average skin color rating from two raters was used as the primary covariate, and its estimated coefficient was the main parameter of interest. To account for unobserved heterogeneity, the model included random effects for player and referee, as well as the average skin color rating across referee's country of origin. While Group 5's analysis illustrates one possible approach, other groups may adopt different data preprocessing decisions, distributional assumptions (e.g., linear, logistic, or Poisson models), specification of fixed and random effects, or conduct the analysis in a Bayesian framework.

We use LLM to convert the methodology description into the schema we propose and generate the corresponding code script. We observe that most reports are structured to first describe the initial data analysis and then the changes made in the final analysis. To avoid testing LLMs' ability to reconstruct how analysis steps are modified, overturned, or carried over from the initial to the final analysis, we extract the descriptions relevant to the final analysis.

We send the text description of each team for LLM to generate the schema and then generate the code script from the schema. The code script is run by a separate LLM to validate for potential syntax errors.

To accelerate the evaluation of each script, we manually in-place replace the `glm()` command with `fixest::feglm()` (Bergé et al., 2026) for generalized linear model and the `lme4::glmer()` (Bates et al., 2015) function with `glmmTMB::glmmTMB()` (Brooks et al., 2017) for generalized mix-effect model after the scripts are generated but before the validation. These two functions are in-place replacements of the standard functions for

estimating the relevant model with multi-threading in C++ (Bergé et al., 2026) and Template Model Builder (TMB) (McGillycuddy et al., 2025) for better performance. This replacement should not affect the estimation of the effect size. All the LLM-related simulations are conducted with claude-opus-4.5.

Across the 25 teams we reproduce, all team scripts were successfully executed after the validation phase. Notably, two scripts run successfully without any validation whereas the vast majority reached completion after two or three iterations. The most frequent correction involved resolving broken and overflow syntax in comments.

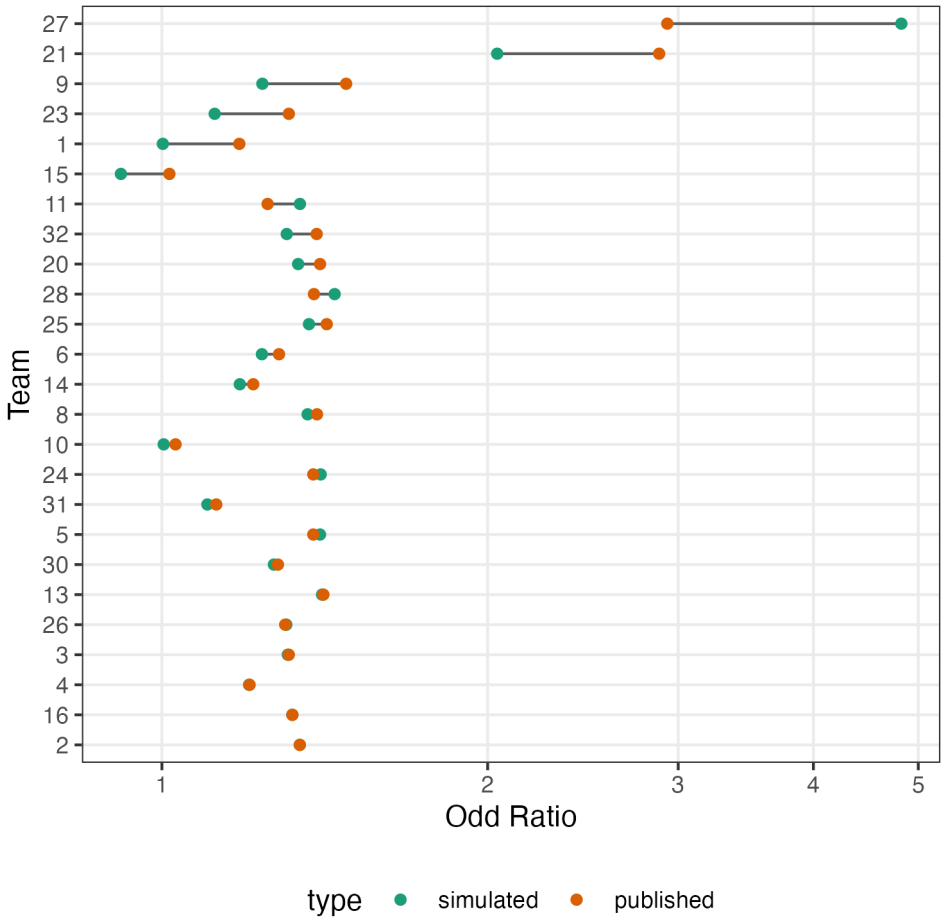


Figure 2. Comparison of the odd ratio of the effect skin color on receiving red cards in football games from our simulation with the schema (simulated) vs. the reported estimated effect size in Silberzahn et al. (2018) (published). Teams are arranged by the difference in odd ratio of our simulation and the reported value. For most teams, the simulation successfully recovers the odd ratio reported by each analyst team, larger variations are seen in the following teams: Team 1, 9, 15, 21, 23, and 27.

Figure 2 shows the comparison of the estimated effect size, in odd ratio, from our workflow (simulated) and those reported in Silberzahn et al. (2018) (published). Our workflow successfully recovers the odd ratio reported in most teams, although larger variation is

observed in the following teams: Team 1, 9, 15, 21, 23, and 27. We further investigate the reports and generated scripts for these teams and find that the discrepancy arises from a combination of ambiguities from the report and LLM's accuracy.

- **Team 27**: the difference comes from the ambiguity of the method description, the authors state that "In addition to the independent variables specified in the initial approach, ... a variable for player position was included, as well as a variable for the league of the player." This description could be interpreted as a shorthand for extending the model with the interaction of additional covariates as how other covariates enters the model in the initial approach, but the LLM adopts a literal interpretation with only the main effect.
- **Team 9**: the authors describe "referee country" in the text as a covariate for the model, but the reported estimation table suggests they use "player country" instead.
- **Team 1 and 23**: the discrepancy comes from the LLM fails to properly reshape the data from player-referee pair level to game level, which requires disaggregating the win, loss, and draw count into individual game records. The schema properly identifies this step but the generated code doesn't execute this step properly and create a silent error with the validator.
- **Team 15**: the discrepancy comes from the default setting for estimating factor variable in R (our simulation) and SPSS (original script). In SPSS, the factor variables are estimated with effect coding, where the coefficients are compared to the grand mean, rather than to the reference group (dummy coding) in R. After changing the `contrast` option in R, we obtain the reported odd ratio.

3.1. *Exploring alternatives*

During the modelling process, researchers may consider a range of alternative specifications, such as different covariate and random effects structures, before arriving at a final model. This can be seen as a set of alternative decisions within the same modelling step, which fits naturally into the schema workflow described in Section 2.2. As an example, we use Team 5 for the above football study to demonstrate the use case. Team 5 uses a binomial generalized linear mixed model (GLMM) to estimate the effect of skin tone rating on the number of red cards received using a logit link function, but tests different random effect structures:

Model 1 (gm1):
$$\begin{aligned}\text{logit}(p_{ijk}) &= \beta_0 + \beta_1 X_{ijk} + u_j + v_k \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}})\end{aligned}$$

Model 2 (gm2):
$$\begin{aligned}\text{logit}(p_{ijk}) &= \beta_0 + \beta_1 X_{ijk} + u_j + v_{0k} + v_{1k} X_{ijk} \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad \mathbf{v}_k \sim N\left(\mathbf{0}, \mathbf{\Sigma}_{\text{ref}}\right)\end{aligned}$$

Model 3 (gm3):
$$\begin{aligned}\text{logit}(p_{ijkm}) &= \beta_0 + \beta_1 X_{ijkm} + u_j + v_k + w_{0m} + w_{1m} X_{ijkm} \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}}), \quad \mathbf{w}_m \sim N\left(\mathbf{0}, \mathbf{\Sigma}_{\text{country}}\right)\end{aligned}$$

where p denotes the probability of a player receiving a red card, and X represents the player's skin tone rating. The subscripts i, j, k, and m index the individual observation, player, referee, and the referee's home country, respectively. The parameter β_0 represents the global baseline log-odds of receiving a red card, and β_1 estimates the fixed effect of a player's skin tone. The random effect include

- a random intercept for players, $u_j \sim N(0, \sigma^2_{\text{player}})$, in Model 1, 2, and 3,
- a random intercept for referee, $v_k \sim N(0, \sigma^2_{\text{ref}})$, in Model 1 and 3, an additional random sloope for referee in Model 2, and
- a random intercept and slope for referee countries, $\mathbf{w}_m \sim N\left(\mathbf{0}, \mathbf{\Sigma}_{\text{country}}\right)$, in Model 3.

These alternatives are specified in a separate YAML file described in Section 2.2 as Listing 3. This file is then merged with the original schema to produce one separate schema for each alternative for code generation and validation. The validated scripts are then executed to obtain the estimated random effects in Table 1. This result matches with the random effects reported in the team report, subject to rounding errors. Similarly, the model fitting statistics can be extracted from the estimation for selecting the best fit.

Listing 3 Example of a YAML file specifying alternatives for the random effects structure in the many-analyst analysis of Group 5.

```
meta:
  type: alternatives
  step: specify_random_effects_structure
  fork: how to account for the non-nested cross-classified data structure
alternatives:
  - id: "gm1"
    path: "a random effect of player and referee"
    rationale: ""
  - id: "gm2"
    path: "a random effect of player and referee, and a random effect
      of skin-tone across referees"
    rationale: ""
  - id: "gm3"
    path: "a random effect of player, referee, and referees' countries
       of origin, and a random effect of skin tone across referees'
      countries of origin"
    rationale: ""
```

Table 1. Summary of random effect estimates for Group 5. The three models (gm1, gm2, and gm3) represent different specifications of the random effects structure to account for the non-nested cross-classified data structure.

Group	Term	gm1	gm2	gm3
Player	Intercept	0.3737	0.3738	0.2899
Referee	Intercept	0.1528	0.1932	0.0748
Referee	Referee rating	NA	0.0286	NA
Referee's country	Intercept	NA	NA	0.3990
Referee's country	Referee rating	NA	NA	0.0870

4. Discussion

Our results highlight a persistent gap between computational reproducibility and analytical understandability in modern data analysis workflows. While current reproducibility practices ensure that published results can be regenerated from shared data and code, as analyses grow in complexity and involve iterative decision-making processes, they are less of an ideal for readers to understand. Our proposed schema addresses this gap by introducing an intermediate representation that explicitly encodes analytical decisions to bridge between narrative descriptions and implementation details. When coupled with LLMs, this representation enables the extraction of analytical structure from written reports and its translation into analysis scripts, and supporting the exploration of alternative analytical choices.

To further examine how this workflow behaves in practice, we apply it to the many-analysts football study. This analysis allows us to identify errors in the original reports and to characterize failure modes of current LLMs to generate code for data science. In particular, we find that LLMs requires a fairly precise specification in order to generate the code correctly for the intended random effect structure. For an educated statistician, the phrase "a random effect of referee number across average rating" would typically imply both a random intercept and a random slope. However, we observe that LLMs interpret this specification literally as a random slope only. This suggests the gap between the latest models and human analysts in specialized statistical knowledge (Sun et al., 2026).

Across the validation process, the LLM successfully corrects a range of implementation issues, including errors caused by overflowing comments and other syntax issues. However, we also observe challenges in data reshaping tasks when the transformation requires changes in the unit of analysis. In the study, some teams decide to transform the data from the original aggregated player-referee dyad-level format to game-level format. This transformation requires disaggregating aggregated outcome counts (win, loss, and draw) into individual game records, changing the unit of analysis from dyads to matches. We also observe some difficulties in parsing birthdates. Since the LLM does not have access to the underlying raw data, due to privacy constraint, it must rely on trial and error when inferring the correct date formats ("DD-MM-YYYY."). This suggests that certain types of data analysis tasks may not be fully automatable in settings where the model cannot directly inspect the data, even if the data is theoretically available for computation.

The reproducibility-understandability issue considered in this work is still at an early stage, some software decision choices are worth examining. We currently implement a prototype as an R package, although the understandability problem is broadly applicable to statistical analyses and is not associated to any particular programming language. Hence, we envision the implementation eventually evolving into a general purpose tool, such as a command-line utility, that can support analyses across different programming languages. In the current implementation, analytical decisions are recorded in a single YAML file. An alternative would be to record individual decisions in separate Markdown files. While YAML files provide an explicit structure for machine readability, markdown files offers greater flexibility for documenting decisions. Recording decisions in separate files would also allow files to be linked across time to understand how decisions evolve or change over the course of an analysis and can potentially support a broader research-management system iattributes decisions to individual researchers and maintains a record of the changes leading to the final analysis.

5. Acknowledgments

The implementation of the schema is available as an R package, `tines`, available on CRAN. The source code for reproducing the work reported in this paper is available in the Supplementary Materials and can be found at: https://github.com/huizezhang-

sherry/paper-schema. The article is created using Quarto (Allaire et al., 2026) in R (R Core Team, 2026).

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